

PROJECT DOCUMENTATION

Meta Ad Performance Analysis — Power BI

End-to-end methodology: data model, SQL exploration, DAX measures, and dashboard design



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Repository	Meta-Ad-Performance-Analysis-PowerBI
Tools used	SQL, Power BI Desktop, DAX, Power Query
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1. Project Overview

This project analyzes advertising performance across Meta's two platforms — Facebook and Instagram — using a synthetic-but-realistic dataset covering 50 campaigns, 200 ad creatives, 400,000 individual ad interaction events, and 9,841 users. The goal is to turn raw event-level logs into a decision-ready Power BI report that a marketing or growth team could use to compare platform performance, judge which ad formats and audiences convert best, and check whether campaign budget is actually buying better results.

400K	50	200	9,841	\$2.5M
AD EVENTS	CAMPAIGNS	AD CREATIVES	USERS	TOTAL BUDGET

1.1 Business Questions

1. Which platform — Facebook or Instagram — delivers a higher click-through and conversion rate?
2. Which ad format (Video, Image, Carousel, Stories) performs best, and does that answer change by platform?
3. Does the ad's targeted gender/age line up with who actually clicks and converts?
4. Does a bigger campaign budget reliably buy a better conversion rate?
5. Are there meaningful time-based patterns (day of week / time of day) worth scheduling around?

2. Data Sources & Schema

Four CSV extracts feed the model. `ad_events` is the fact table; `ads`, `campaigns`, and `users` are dimension tables joined to it.

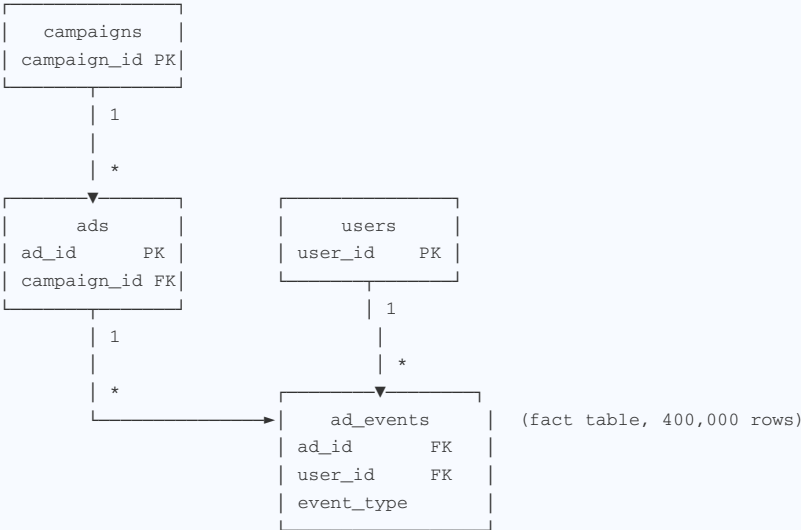


Table	Rows	Grain	Role in model
campaigns	50	1 row / campaign	Dimension — budget & flight dates
ads	200	1 row / ad creative	Dimension — platform, format, targeting
users	9,841	1 row / user	Dimension — demographics
ad_events	400,000	1 row / interaction	Fact — impressions through purchases

Full column-level definitions live in **Data_Dictionary.xlsx**, alongside this document.

3. Methodology

3.1 Data Cleaning & Quality Checks (SQL / Power Query)

Before modeling, each table was profiled for nulls, duplicate keys, and referential integrity between the fact table and its dimensions (see `SQL Queries/Data_Exploration.sql`). Two real issues were found and should be corrected upstream:

Data quality issue — corrupted user IDs. `users.csv` uses 5-character alphanumeric codes as `user_id`. Roughly 1,240 of the 9,841 IDs (12.6%) happen to be all-digit codes (e.g. `00012`), and were silently reformatted by Excel into numbers / scientific notation (`1.20E+01`) the last time the file was saved — stripping leading zeros. This breaks the join to `ad_events.user_id` for any row that used one of the affected IDs: **455 of the 9,950 distinct users in ad_events (4.6%) cannot be matched back to a demographic profile** as a result.

Recommended fix: re-export `users.csv` with `user_id` forced to a text column (or zero-padded), and in Power Query set the column type to

Text with "Do not detect data types" enabled on import, so this cannot recur.

Date format caveat. `campaigns.start_date` / `end_date` are stored as text in `DD-MM-YYYY` format (e.g. `25-05-2025`). Power BI's auto-detect defaults to US `MM-DD-YYYY` and will silently swap day and month for any date where the day is ≤ 12 , corrupting roughly 40% of rows without raising an error. Dates must be parsed with an explicit `DD-MM-YYYY` format or an `en-GB` locale set in Power Query.

Coverage caveat. `ad_events` timestamps run 07-May-2025 to 06-Aug-2025 (≈ 3 months), while campaign flight dates span 13-Feb-2025 to 12-Oct-2025. Campaigns running outside the May-Aug window have partial or no interaction data in the fact table — treat cross-campaign comparisons with that in mind, and don't read "zero events" as "zero performance" for those campaigns.

Other checks came back clean: no duplicate primary keys in any table, no null values in any column, and every `ad_id` / `campaign_id` foreign key in the fact and bridge tables resolves to a real parent row.

3.2 SQL Analysis

Script	Purpose
<code>Data_Exploration.sql</code>	Row counts, null checks, distinct-value profiling, referential integrity checks across all four tables
<code>KPI_Analysis.sql</code>	Aggregate CTR, engagement rate, conversion rate and purchase rate, overall and by platform
<code>Ad_Performance_Analysis.sql</code>	Campaign- and ad-type-level performance ranking, including cost-per-click and cost-per-purchase

3.3 Power BI Data Model

The four tables are loaded into Power BI and related in a star-schema shape: `ad_events` is the single fact table, related many-to-one to `ads` (on `ad_id`) and to `users` (on `user_id`); `ads` is related many-to-one to `campaigns` (on `campaign_id`). All relationships are single-direction filter flow from dimension to fact, which is what lets slicers on platform, ad type, gender, and country filter the 400,000-row fact table efficiently.

3.4 DAX Measures

Thirteen measures were built to power the cards, charts, and the ad-type performance table. The full list — including the exact logic behind the "Engagements" figure, which combines Clicks, Comments and Shares rather than every interaction type — is documented in the **Measures** sheet of `Data_Dictionary.xlsx` .

3.5 Dashboard Pages

Page	What it shows
Overview / Clicks	KPI cards (Impressions, Clicks, Shares, Comments, Purchases, Engagements, CTR, Engagement Rate, Conversion Rate, Purchase Rate, Total Budget); clicks by gender and age; weekly and hourly click trend; clicks by country map; a calendar heatmap; an ad-type performance breakdown table. Filterable by Facebook/Instagram and a Month slicer.
Comments	The same layout, retargeted at Comments as the measured interaction instead of Clicks — comments by gender/age, weekly/hourly comment trend, comments by country.

Both pages share a right-hand rail with Meta/Facebook/Instagram platform branding and slicers for ad name and a dynamic measure selector, so the same visuals can be repointed at different metrics without duplicating pages.

4. Assumptions & Limitations

- **No revenue field.** "Purchases" is an event count, not a dollar value — the dashboard measures purchase volume and rate, not ROAS or revenue. Adding a `purchase_value` field to `ad_events` would be the natural next step for a true ROI view.
- **Total Budget is not platform-split.** Because a single campaign can run ads on both Facebook and Instagram, the `Total Budget` and `Avg. Budget per Campaign` cards show the same figure regardless of the platform slicer. A platform-level budget view would require allocating each campaign's budget across its ads first (e.g. evenly, or by impression share).
- **"Engagement" excludes Likes.** The Engagements measure sums Clicks + Comments + Shares only. This is a deliberate modeling choice worth flagging in any write-up, since most teams' informal definition of "engagement" also includes Likes.
- **Targeted vs. actual audience are different fields.** Gender/age charts on the dashboard are built from what the *ad was targeted at* (`ads.target_gender` , coarse 4-band age), not who actually clicked (`users.user_gender` , finer 6-band age). Both are useful, but they answer different questions and shouldn't be read interchangeably.
- **Time-of-day and day-of-week buckets are close to uniform** in this dataset (each weekday and each of the four day-parts account for roughly the same share of events), which limits how much can be concluded about optimal scheduling from this data alone.